

- 5 Yeast cells respond to changes in glucose concentration in their environment by using transcription factors to switch off genes.

When glucose is **present**:

- Mig1 transcription factors bind to the promoters of five genes
- Mig1 binding to the promoters stops (represses) transcription of these genes.

The genes that are repressed by Mig1 code for five enzymes that allow yeast cells to metabolise the sugar galactose when glucose is absent.

- (a) Complete Table 5.1 to show **three chemical** differences between a transcription factor, such as Mig1, and a promoter.

Table 5.1

	transcription factor	promoter
difference 1		
difference 2		
difference 3		

[3]

(b) Mig1 binds to promoter sites with these features:

- 17 base pairs long
- includes a region of five repeating adenine-thymine pairs
- includes a region of six repeating cytosine-guanine pairs.

Promoter sites to which Mig1 binds are known as Mig1-binding promoter sites.

Bioinformatic techniques were used to analyse the yeast genome to look for sections of DNA that match these features. The information obtained for four chromosomes is shown in Table 5.2.

Table 5.2

yeast chromosome	chromosome size /base pairs	number of Mig1-binding promoter sites per chromosome
A	230 018	1
B	813 184	10
C	316 620	2
D	1 531 933	14

(i) Explain why bioinformatic techniques were used to obtain the information in Table 5.2.

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..... [2]

(ii) Identify, with a reason, the yeast cell chromosome that is **most likely** to include genes that code for enzymes that metabolise galactose.

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..... [1]

- (iii) Mig1 binds to 27 promoters on these four chromosomes. Yeast cells also have other chromosomes where Mig1 binds to additional promoters.

Five different enzymes, coded by five genes, must be made for yeast cells to metabolise galactose.

Suggest reasons why an individual diploid yeast cell has a larger number of Mig1-binding promoter sites than the expected number of ten.

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..... [2]

- (c) The repression of genes involved in galactose metabolism in yeast is similar to events at the *lac* operon in the bacterium *Escherichia coli*.

Explain how *E. coli* **represses** the production of proteins needed to metabolise lactose sugar.

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..... [3]

[Total: 11]